Evals for AI catalog

Objective

To design and deploy a robust enrichment evaluation and approval system using multi independent AI models. The system will:

- Standardize evaluation formats across models (match types + reasons)
- Automatically approve exact matches with high confidence (Model A only)
- Route all other results to a human review interface
- Provide distinct model outputs for comparison
- Offer visual clarity via color-coded match indicators and confidence bars (Model A only)
- Log all decisions for traceability and model feedback

Background

CC uses two distinct AI models for product catalog enrichment. These models analyze product SKUs and suggest missing or updated attribute values. Each model outputs evaluations, but with differing formats and capabilities:

- Fynd's Catalog Classification (CC) team uses two enrichment models:

	Model	Match Types	Confidence Score	Notes
1	**Model A**	✅ Yes	❌ No	Internal model
2	**Model B**	✅ Yes	✅ Yes	Gemini model (Confidence score- to be confirmed by Google)

Both models now output standardized **evaluation types**, e.g.:

- `exact`, `similar`, `partial`, `not_similar`, `no_data`, `failed`
- Along with a `reason` for match

However, only **Model B** provides attribute-level **confidence scores**, which will be used to guide **auto-approvals**.

Previously, enrichment decisions were guided by **SKU Health**, incorporating confidence scores. With the newer model outputs, there is a need to:

- Automate approvals where results are deemed reliable - `exact match`
- Introduce a **human-in-the-loop (HITL)** pipeline for ambiguous or low-confidence results-`partial`, `not_similar`, `no_data`, `failed`

Scope

IN-SCOPE:

- Evals rule has to be on source: Model A, Model B
- Display predictions, match types, reasons from both models

- Show Model B confidence score
- Allow reviewers to:
  - Compare both predictions
  - Approve either model's output
  - Manually override
- Color-coded UI based on match type
- Excel export/import support - with color coding
- Logging of all decisions with reviewer metadata

OUT-OF-SCOPE (CURRENT PHASE):

- Auto-Approvals based on the Confiodence scores + SKU Health
- Workflow management

Input Schema Overview

1. GEMINI schema:

```
{ "request_uuid":"", "channel":"", "sku":"", "l1_category":"", "l2_category":"", "l3_c
```

2. Internal FYND schema:

```
{ "optioncode": "", "sku": [], "l1_category": "", "l2_category": "", "l3_category": "'
```

Decision Logic

AUTO-APPROVAL RULES

	Criteria	Action	C
1	Model A/B: `match_type == exact`	✅ Auto-approve attribute	if all attributes are auto approved then SKU is to be marked Approved?
2	All other types	Human evaluation (HTL)	Can override

Human Override Behavior

When a human edits or overrides:

- Final value replaces both model predictions—where there's `exact match type`
- **A tag (`Human Override`) is shown -** Tooltip: “Evaluation — manually corrected”
- This is even to be done when manually overridden by Excel / UI

Color Coding & UI Rules

MATCH TYPES:

	Match Type	Color	Icon
1	Exact	Green	✅ (Green in excel as well)
2	Partial Match, Not Similar, No Data	Red	❌ (Show red color in excel as well) - Shriyash Jadhavshare # code

EXPORT COLUMNS:

- SKU
- Attribute
- Model A prediction + match + reason
- Model B prediction + match + reason
- Final value
- Decision source (Model A / B / Manual)

IMPORT FORMAT:

- Final decision overrides in bulk
- Validated input structure and reviewer metadata

Open question:

1. If both models results are kept seperate - how do you finally share a value to downstream- say Attribute A i approved from Model A, Attribute B from B -?
2. if all attributes are auto approved then SKU is to be marked Approved?
3. What happens to confidence score / SKU health on edit

Product Requirements Document (PRD)

Feature: Multi-AI Model Evaluation & Approval Workflow in Catalog Cloud

1. OBJECTIVES

- Efficiently manage product enrichment data from multiple AI models.
- Enable precise human-in-the-loop (HITL) validation and audit of AI-generated catalog attributes.
- Provide clear, auditable decision trails for regulatory and operational compliance.

2. FUNCTIONAL OVERVIEW

Catalog Cloud incorporates AI enrichment data from two models (Model A and Model B), which provide attribute predictions, confidence scores, match types, and reasons.

3. DATA MANAGEMENT

Attribute-Level AI Data:

- **Prediction:** Suggested value from AI.
- **Match Type:** exact, similar, partial, not_similar, no_data, failed.
- **Confidence Score:** Numeric (0.0 to 1.0).
- **Reason:** Explanation from AI for prediction or evaluation.

4. AUDIT AI DATA PAGE

Workflow:

- **View Multi-Model Predictions:** Display attributes enriched by Model A & B, showing predictions, match type, and confidence.
- **Bulk Selection & Actions:** Approve, Reject, Partial approval via checkboxes and bulk-action toolbar.
- **Individual Row Actions:** Approve or reject individually; explicitly mark "partial" approvals for mixed evaluations.

Decision Logic:

- **Approve:** Accept AI prediction; store final approved value.
- **Reject:** Discard AI predictions, preserve existing catalog value.
- **Partial:** Accept some attributes; explicitly mark others for manual review or reject.

5. SKU DETAILS PAGE

Workflow:

- **View Attribute Details:** Display AI-suggested values alongside existing attribute values.
- **Toggle AI-Enrichment View:** Enable/disable the visibility of AI enrichment suggestions.
- **Approve/Reject/Partial:** Actions specific to each attribute.

Decision Handling:

- **Approve:** AI value overwrites existing catalog value; marked as "AI-Approved."
- **Reject:** Existing catalog value retained; AI suggestion logged as rejected.
- **Partial Approval:** Manually choose or edit individual attributes; track and audit these manual interventions.